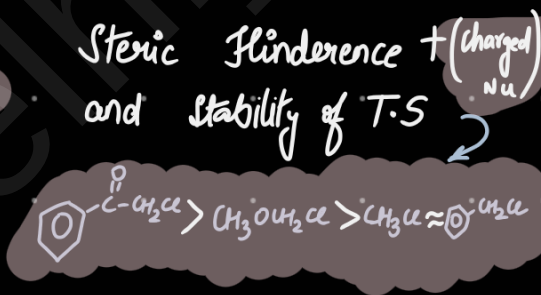
SUBSTITUTION NUCLEOPHILIC

REACTIONS MECHANISMS

	S_N1	S_N2
1. Rate	$R = K[subs]^1$	$R = K[subs]^1[Nu]^1$
2. Molecularity	1	2
3. Order	1	2
4. Favoured Subst.	$3^\circ > 2^\circ > 1^\circ > CH_3X$	$CH_3X > 1^\circ > 2^\circ > 3^\circ$
5. Feasibility	Stability of $C^\oplus$ + (weak Nu) ↳ 1. Aromatic 2. Resonance 3. Hyperconjugation 4. Inductive	Steric Hinderence + (charged Nu) and Stability of T.S  * In 2° subs → Nu like $\{CN^\ominus, SH^\ominus, N_3^\ominus, I^\ominus\}$ → S_N2
6. RDS	Formation of $C^\oplus$ (first)	Single Step rxn.
7. Hybridisation	$sp^3 \rightarrow sp^2 \rightarrow sp^3$	$sp^3 \cdots sp^2 \cdots sp^3$
8. Stereochemistry	Racemization (inversion ≥ retention)	Inversion of config.
9. Solvent	Polar Protic eg. R-OH, H_2O, NH_3, R-C(=O)-OH (solvation of the ions formed)	Polar Aprotic eg. DMSO, $CH_3-C(=O)-CH_3$ (acetone), DMF, THF, HMPT * If subs and Nu both are neutral ⇒ protic
10. To be paid attention	$C^\oplus$ rearrangement	NGP

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11. Nucleophilicity

(I) Polar Aprotic ⇒ **Nu = Basicity**

eg. $F^- > Cl^- > Br^- > I^-$

$^-OH > ^-SH$

Polar Protic ⇒ **Nu = Polarizibility**

eg. $I^- > Br^- > Cl^- > F^-$

eg. $^{\ominus}SH > ^{\ominus}OH$

(II) In a period ⇒ Nu decreases (∵ Both basicity and polarizibility are same)
(Both protic + aprotic) eg.- $^-CH_3 > ^-NH_2 > ^-OH > ^-F$

(III) C.B > C.A eg. $^-OH > H_2O$

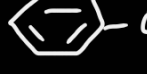
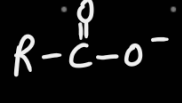
★ $^-OCH_3 > ^-OH$

(IV) Nu site is attached to an atom with LP ⇒ Nu ↑↑

eg. $H_2\ddot{N}-\ddot{N}H_2 > \ddot{N}H_3$

$H\ddot{O}-O^{\ominus} > HO^{\ominus}$

(V) Same Nu atom ⇒ Nu = Basicity

$^-OH >$  $>$ 

(VI) **Crowdness** $\alpha \frac{1}{Nu}$ eg. $CH_3S^- > (CH_3)_2CHS^-$

(although this is a better base)

When the compounds belong to the same family ⇒ Basicity ∝ Crowdedness ✓

Ambident Nu: 2 Nu sites.

eg. $RX + NaC\ddot{N} \rightarrow RCN + NaX$

$RX + AgC\ddot{N} \rightarrow RNC + AgX$

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12. SHAB concept :

L. Base → Soft : e^- pair on large size atom = Less magnitude of $R^{+\delta}$ (S_N2) +ve charge : Soft ← L. Acid

L. Base → Hard : e^- pair on small size atom = More magnitude of R^+ (S_N1) +ve charge : Hard ← L. Acid

eg. $(CH_3)_3C-Br$ (Hard) S_N1 $\xrightarrow[(C \to Soft,\ N \to Hard)]{NaCN}$ $(CH_3)_3-C-NC$

CH_3-Br (Soft) S_N2 $\xrightarrow{NaCN}$ CH_3-CN

14.

a) 

$\Rightarrow$ S_N1 xxx

S_N2 xxx

b) $CH_3-C(CH_3)(CH_3)-CH_2Cl$ $\Rightarrow$ S_N1 xxx

$\Rightarrow$ S_N2 xxx

↳ (better than 3°, worse than 2°)

c) $S_N1 \Rightarrow$ 

due to angle strain (II) is unstable → quickly react to form 

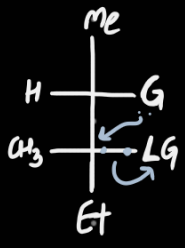
15. NGP

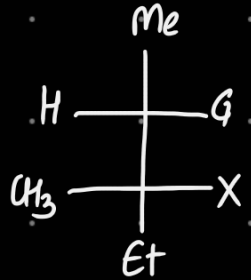
↳ increases the rate many times.

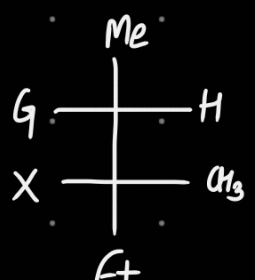
↳ net → retention of configurations

Groups : Sulphide, esters, Carboxylate, $O^\ominus$, Amine, (Br < I), π's. ✱ Ethers xxx

↳ 1-3, or 1-5 attack.

a) 

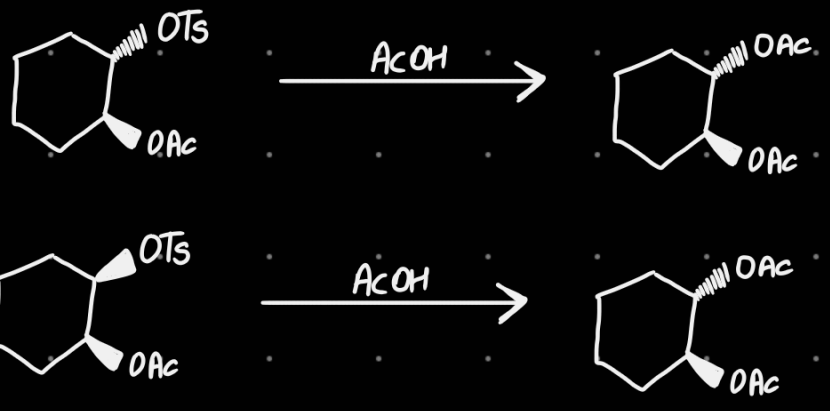
$\xrightarrow{HX}$ 

+ 

dono pe retention

dono pe inversion

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b) 

retention due to NGP

ANTI

inversion ↳ S_N2 NO NGP!

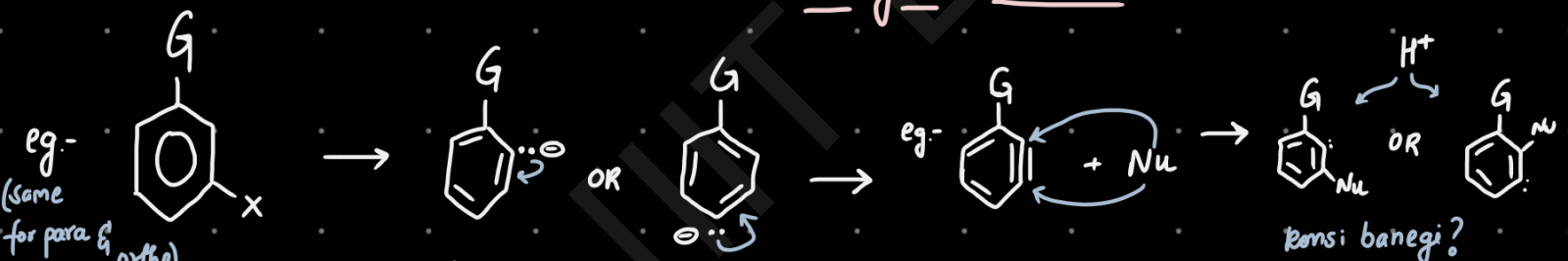
16. S_Ni

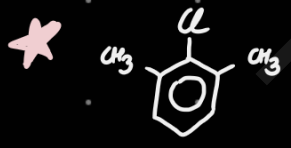
Nu andar hi baitha hai!



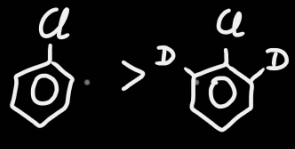
16. ArS_N1 - Elimination Addition

Benzyne Mechanism



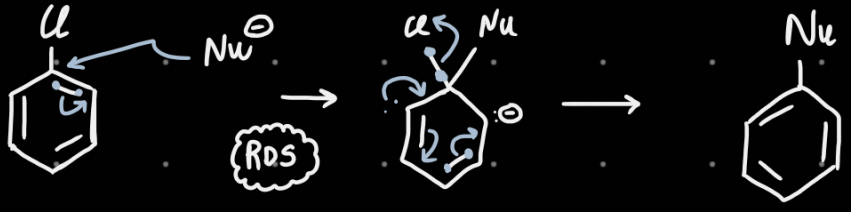
★ 

→ XXX no ortho 'H' ∴ no benzyne.

★ 

★ RDS ⇒ formation of Carbanion.

17. Ar_SN2 - Addition Elimination



(Meisenheimer complex)

★ eWG at ortho para ⇒ rate ↑↑

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